

COMMUNICATION LABORATORY

Experiment 4: Delta Modulation and Demodulation



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TITLE: Design of Delta Modulation and Demodulation scheme

1. PROBLEM STATEMENT:

1. Design, implement, and optimize a Delta Modulation (DM) and Demodulation hardware circuit to convert a continuous analog signal into a 1-bit digital pulse train and reconstruct it, actively mitigating slope overload and granular noise by tuning the sampling frequency (f_s) and integrator RC cutoff frequencies (f_c).
2. Extract the high-resolution modulated and demodulated data streams using a Digital Storage Oscilloscope (DSO), and utilize MATLAB to perform rigorous software-domain analysis to visually verify the microscopic staircase tracking and macroscopic signal recovery.

2. EQUIPMENT LIST & KEY COMPONENTS:

- **IC 710 (High-Speed Differential Comparator):** Used in the modulator to instantly compare the analog message signal $m(t)$ against the integrated staircase prediction $\hat{m}(t)$
- **IC 7474 (Dual D-Type Flip-Flop):** Acts as the 1-bit quantizer and discrete time sampler, triggered by the 80 kHz clock to produce the digital pulse train.
- **IC 741 (Operational Amplifier):** Configured as a high-speed level shifter (comparator) to convert the unipolar 0-5V TTL output of the flip-flop into a symmetrical bipolar signal.
- **IC 356 / LF356 (JFET Operational Amplifier):** Used in the demodulator for active, high-input-impedance integration to prevent capacitor charge leakage.
- **Function Generator:** Used to provide the analog message signal ($f_m = 1$ kHz, $5V_{pp}$) and the high-speed sampling clock ($f_s = 80$ kHz).
- **Digital Storage Oscilloscope (DSO):** Used for real-time waveform visualization and exporting massive data arrays to CSV format.
- **Passive Components (Tuned for optimal $f_c = 4$ kHz tracking):**
 - **Level Shifter Voltage Divider:** $R1 = 1$ k Ω , $R2 = 1$ k Ω (generates the $V_{ref} = 2.5$ V threshold).
 - **Modulator RC Integrator:** Assorted test values used during our full tuning

sweep (from 16 kHz down to 1.5 kHz):

- Test 1: 1 k Ω +10 nF (yields $f_c \approx 15.9 \text{ kHz} \approx 16 \text{ kHz}$)
- Test 2: 1.5 k Ω +10 nF (yields $f_c \approx 10 \text{ kHz}$)
- Test 3: 10 k Ω +5.6 nF (yields $f_c \approx 2.8 \text{ kHz} \approx 3 \text{ kHz}$)
- Test 4: 10 k Ω +8.2 nF (yields $f_c \approx 1.9 \text{ kHz} \approx 2 \text{ kHz}$)
- Test 5: 5.6 k Ω +0.02 μ F (yields $f_c \approx 1.4 \text{ kHz} \approx 1.5 \text{ kHz}$)
- Final optimized components equivalent to exactly $f_c = 4 \text{ kHz}$.
- **Demodulator Active Integrator:** $R = 5.6 \text{ k}\Omega$, $C = 6.8 \text{ nF}$.
- **Demodulator Low Pass Filter:** $R = 15 \text{ k}\Omega$, $C = 10 \text{ nF}$ (yielding a cutoff of approx 1 kHz).
- **Breadboard, Connecting Wires, and Power Supply (+12V, -5V, +5V, GND).**

(See Appendix A for IC PIN Diagrams).

3. BRIEF THEORY & BACKGROUND

3.1 The Need for Analog-to-Digital Modulation

In modern communication systems, transmitting raw analog signals is highly susceptible to channel noise, distortion, and attenuation. By digitizing the signal into a binary format, we can employ error correction, digital encryption, and signal regeneration at repeaters. Pulse Code Modulation (PCM) is the traditional method for digitizing analog signals, but it is highly bandwidth-intensive because it transmits multi-bit absolute values (e.g., 8 or 16 bits) for every single sample [1].

Delta Modulation (DM) was invented to solve this exact bandwidth problem. Instead of transmitting the absolute value of a sample, DM transmits only a single bit that indicates the *change* (or delta) from the previous sample.

3.2 The Mathematics of Sampling and Oversampling

According to the Nyquist-Shannon sampling theorem, a signal with maximum frequency f_m must be sampled at $f_s \geq 2f_m$ to prevent aliasing [2]. However, Delta Modulation relies on a concept known as **oversampling**, where the sampling rate is vastly higher than the Nyquist rate ($f_s \gg 2f_m$). By sampling the signal incredibly fast, the voltage difference between any two adjacent samples is guaranteed to be extremely small. This high correlation between adjacent samples is what allows us to encode the entire signal change using only a single bit (+1 for an upward trend, -1 for a downward trend).

Rule of Thumb: As a practical standard in communication design, to effectively solve slope overload and tracking issues, the sampling frequency is typically chosen to be at least 20 times the maximum message frequency ($f_s \geq 20f_m$).

3.3 Deep Dive into the Delta Modulation Architecture

Delta Modulation is effectively a 1-bit Differential Pulse Code Modulation (DPCM) scheme. The architecture relies on three primary blocks:

1. **The Error Detector (Comparator - IC 710):** The modulator maintains a running "guess" of what the signal looks like, known as the staircase approximation $\hat{m}(t)$. The highly sensitive IC 710 differential comparator subtracts this guess from the actual message signal $m(t)$ to generate an error signal $e(t) = m(t) - \hat{m}(t)$ [3]. If the actual signal is higher than the guess, the error is positive, and the comparator output hits its positive saturation voltage. If the actual signal is lower, the error is negative, and it hits negative saturation.
2. **The 1-Bit Quantizer (D Flip-Flop - IC 7474):** The continuous-time error signal must be discretized. The D Flip-Flop is clocked at the high-frequency f_s . On every rising edge of the clock, it latches the comparator's state. This converts the analog error into a strict, discrete-time digital stream of 1s and 0s.
3. **The Accumulator (Integrator):** This is the heart of the feedback loop. The digital 1s and 0s (level-shifted to $+V$ and $-V$) are fed into an integrator. In physics, the integral of a constant positive voltage is a positive-sloping voltage ramp; the integral of a constant negative voltage is a negative-sloping voltage ramp. Thus, a stream of 1s creates a staircase stepping upwards, and a stream of 0s creates a staircase stepping downwards. This staircase becomes $\hat{m}(t)$.

3.4 Deep Dive into DM Distortions

The engineering challenge in Delta Modulation is entirely focused on balancing two conflicting distortions:

- **Slope Overload:** This occurs when the analog message signal $m(t)$ is rising or falling extremely fast. The staircase can only climb at a maximum speed dictated by its step size (Δ) and the sampling frequency (f_s). If the analog signal's slope $|dm(t)/dt| > \Delta/T_s$ (or $\Delta \cdot f_s$), the staircase simply cannot climb fast enough to catch the signal [3]. The output becomes severely distorted. To fix this, you must either increase the step size Δ or increase the sampling clock f_s .
- **Granular Noise:** This is the opposite problem. When the analog signal is relatively flat (like at the peak of a sine wave), the modulator has no "stay flat" option—it must step UP or DOWN. Therefore, the staircase aggressively oscillates above and below the flat signal by exactly $\pm\Delta$. This creates a high-frequency buzz known as idle-channel noise or granular noise [2]. To fix this, you must decrease the step size Δ , which directly contradicts the solution for slope overload.

3.5 Advantages and Real-World Limitations

The brilliance of Delta Modulation lies in its hardware simplicity: it bypasses the need for expensive, complex successive-approximation ADCs. However, because the step size Δ is fixed, it struggles to adapt to signals with highly variable dynamic ranges (like human speech or music). This limitation ultimately led to the invention of Adaptive Delta Modulation (ADM), where the step size Δ automatically expands during steep

slopes and shrinks during flat regions [1].

4. DESIGN & CIRCUIT IMPLEMENTATION

4.1 Modulator Design & Calculations

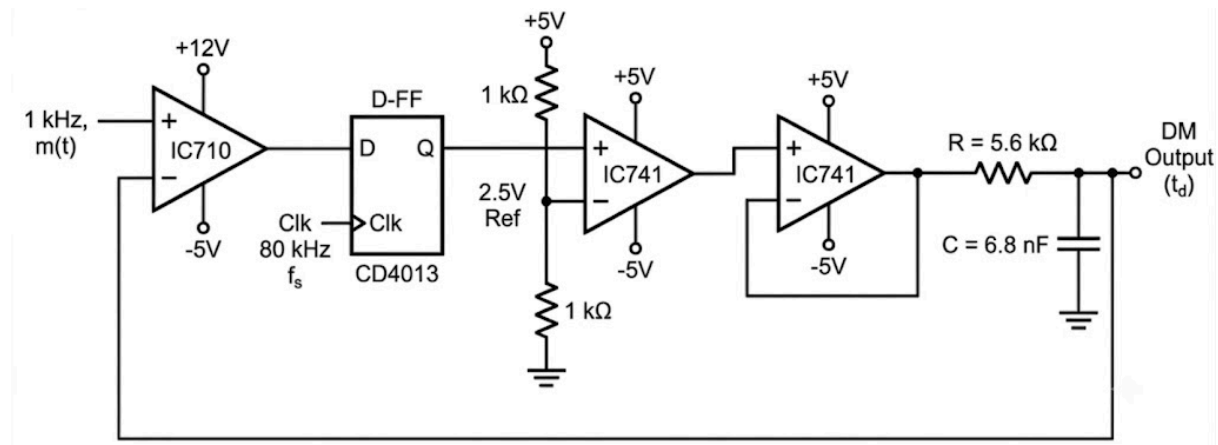


Fig 4.1 - Detailed circuit schematic of the implemented Delta Modulator, featuring the IC 710 comparator, IC 7474 D-Flip Flop quantizer, IC 741 bipolar level shifter, and the buffered passive RC integrator network.

1. **Comparator (IC 710):** Evaluates the analog message $m(t)$ (1 kHz) against the staircase $\hat{m}(t)$.
2. **Quantizer (IC 7474):** Latches the comparator output at exactly $f_s = 80$ kHz.
3. **Level Shifting (IC 741):** The 7474 outputs unipolar 0-5V TTL logic. An integrator fed 0-5V would only ever integrate positively, causing the staircase to fly off to infinity. We must shift this to bipolar logic. By creating a voltage divider ($R_1 = 1$ kΩ, $R_2 = 1$ kΩ) across a 5V supply, we create a 2.5V reference. The IC 741 compares the 0-5V data stream against this 2.5V threshold, perfectly swinging its output from -5V to +5V.
4. **Passive Integrator & Buffer:** Initially, an active op-amp integrator was attempted for the feedback loop. However, active integrators without perfect DC stabilization are highly prone to drift and rail saturation, causing the circuit to fail. We shifted to a robust passive RC integrator to generate the staircase. A voltage buffer was placed after the passive network to prevent loading.
 - **Calculations (Formula for Δ):** The actual step size Δ generated by an RC integrator driven by a bipolar voltage swing V over the sampling period T_s is mathematically given by: $\Delta = (V \cdot T_s) / RC = V / (RC \cdot f_s)$. To completely prevent slope overload, the maximum rate of rise of the staircase must exceed the

maximum derivative of the message signal ($m(t)=A_m\sin(2\pi f_m t)$). The mathematical condition is: $\Delta / T_s \geq |dm(t)/dt|_{\max} \Rightarrow V / RC \geq 2\pi \cdot f_m \cdot A_m$. For our message ($f_m=1$ kHz, $A_m=2.5$ V), the max slope is $2\pi(1000)(2.5)\approx 15,707$ V/s. Using our bipolar swing $V=5$ V, we theoretically require an RC time constant $RC \leq 5/15707 \approx 318 \mu\text{s}$. The corresponding cutoff frequency $f_c=1/2\pi RC$ guided our manual tuning toward the 4 kHz optimal zone.

4.2 Demodulator Design & Calculations

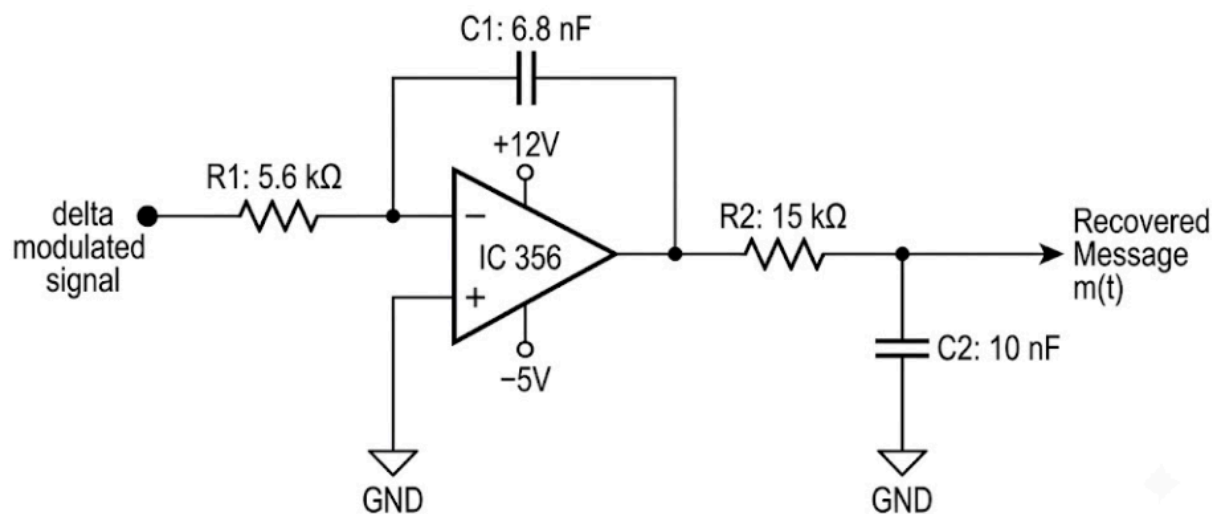


Fig 4.2 - Detailed circuit schematic of the implemented Delta Demodulator, featuring the IC 356 active JFET integrator and the final passive RC Low-Pass Filter.

- **Active Integration (IC 356):** At the receiver, the raw bit stream is pushed through an active JFET integrator ($R = 5.6$ kΩ, $C = 6.8$ nF). The JFET prevents the capacitor from leaking charge, recreating an identical, highly stable staircase.
- **Low Pass Filter:** A passive RC LPF (15 kΩ and 10 nF) smooths out the jagged edges (granular noise). The cutoff frequency is calculated as $f_{\text{cutoff}} = 1 / (2\pi RC) = 1 / (2\pi \cdot 15\text{k}\Omega \cdot 10\text{nF}) \approx 1.06$ kHz. This perfectly strips the high-frequency quantization noise while preserving the 1 kHz baseband message.

4.3 Expected Outputs (Manual References)

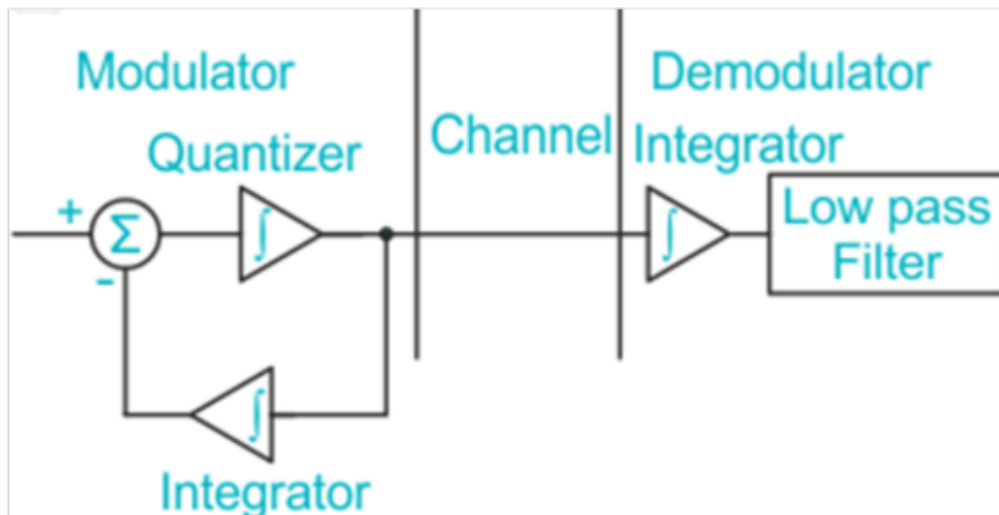


Fig 4.2 - Simplified block diagram of a Delta Modulator and Demodulator.

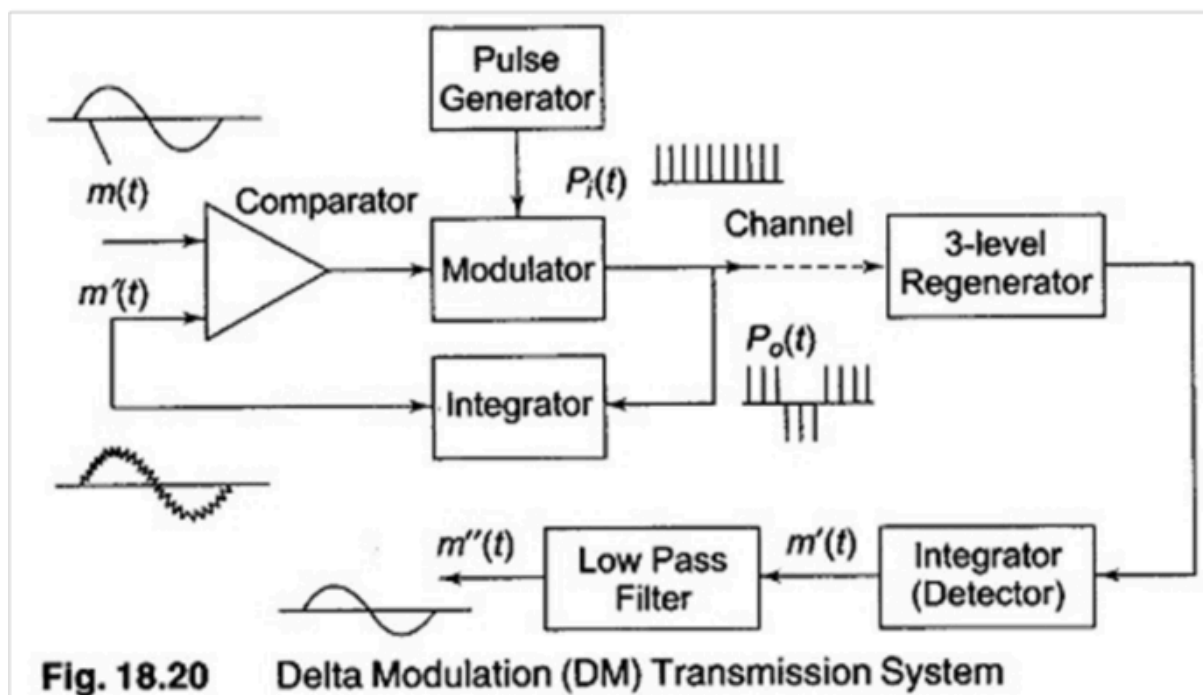


Fig 4.3 - Detailed Delta Modulation (DM) Transmission System showing the expected waveforms at each internal node.

5. EXPERIMENT RESULTS & PLOTS

The physical circuit was rigorously tested on the DSO, and data was extracted to CSV for extensive software validation in MATLAB. Our systematic tuning of the f_c equivalent cutoff was captured through carefully labeled screenshot files.

5.1 Oscilloscope Waveforms (Hardware Results)



Fig 5.1 - Severe Slope Overload. RC components ($0.02\mu\text{F}$ and $5.6\text{ k}\Omega$) yielded an $f_c \approx 1.5\text{ kHz}$. The massive RC time constant caused the step size to be far too small to climb the sine wave.

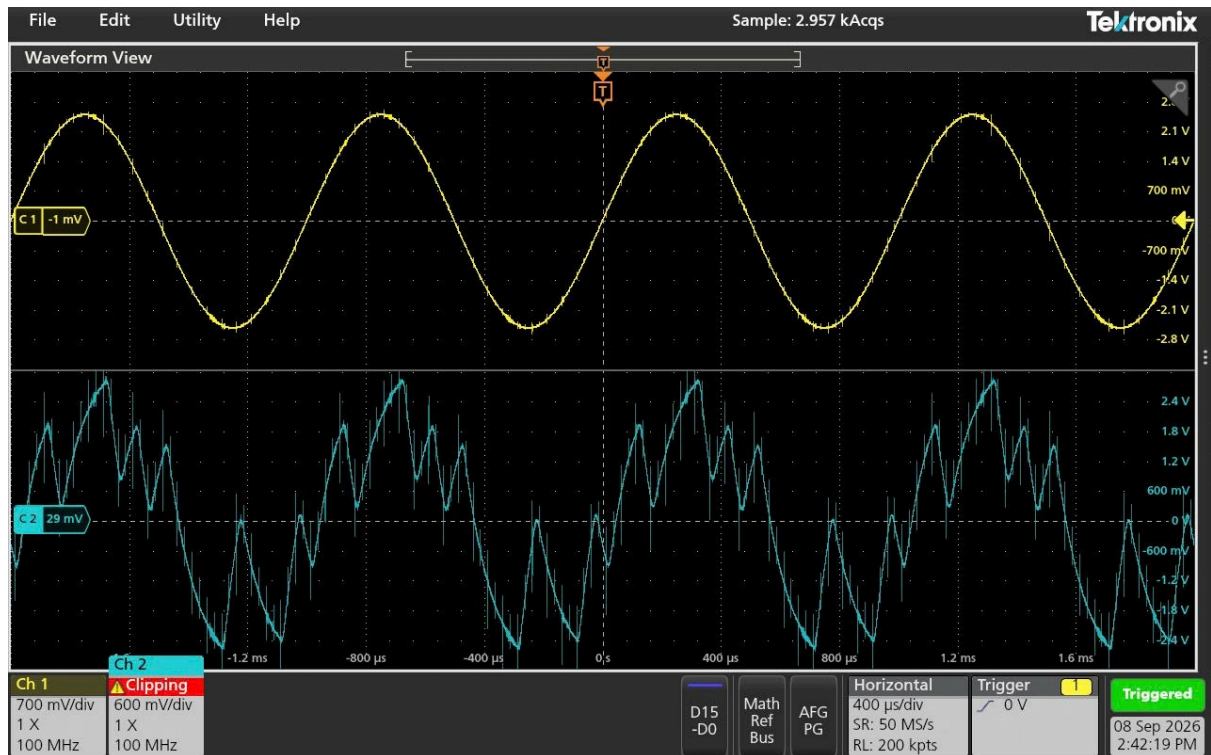


Fig 5.2 - Moderate Slope Overload. RC components (8.2 nF and 10 kΩ) yielded an $f_c \approx 2$ kHz. The staircase attempts to track but still falls short on the steepest gradients.

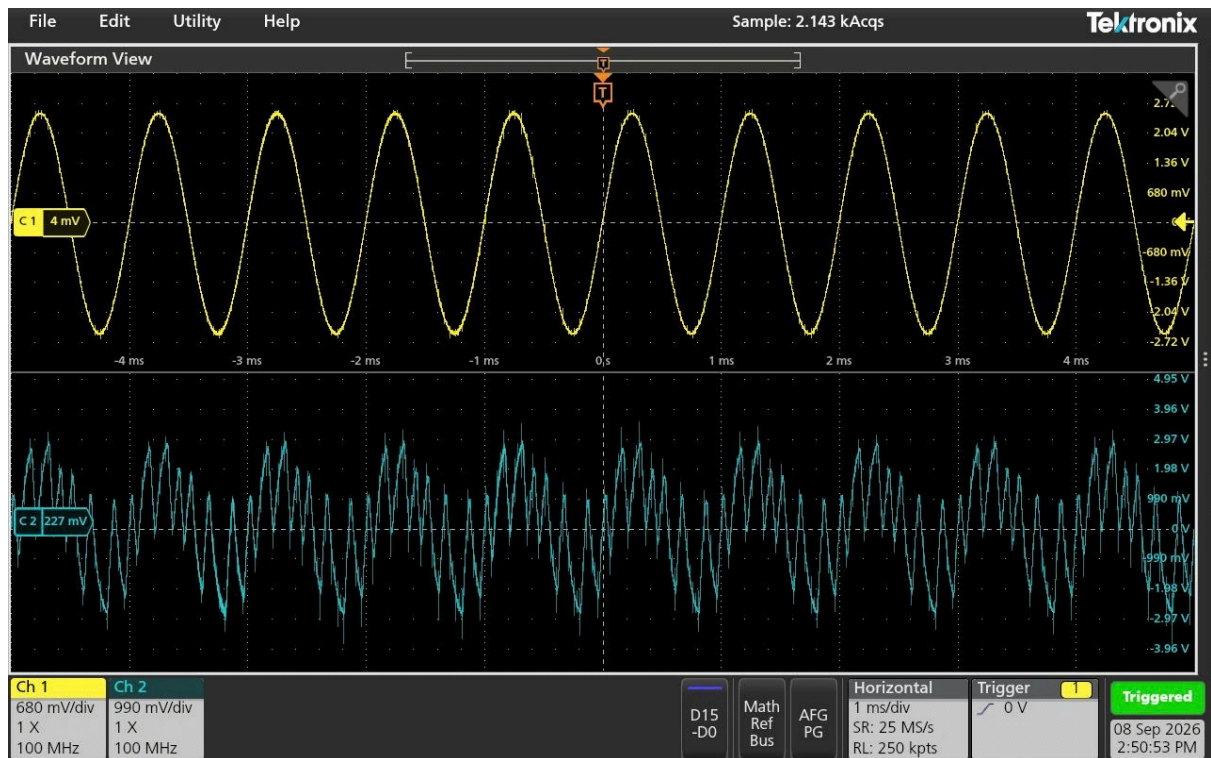


Fig 5.3 - Marginal Tracking. RC components (5.6 nF and 10 kΩ) yielded an $f_c \approx 3$ kHz. Tracking is significantly improved but slight distortion remains on the peaks.

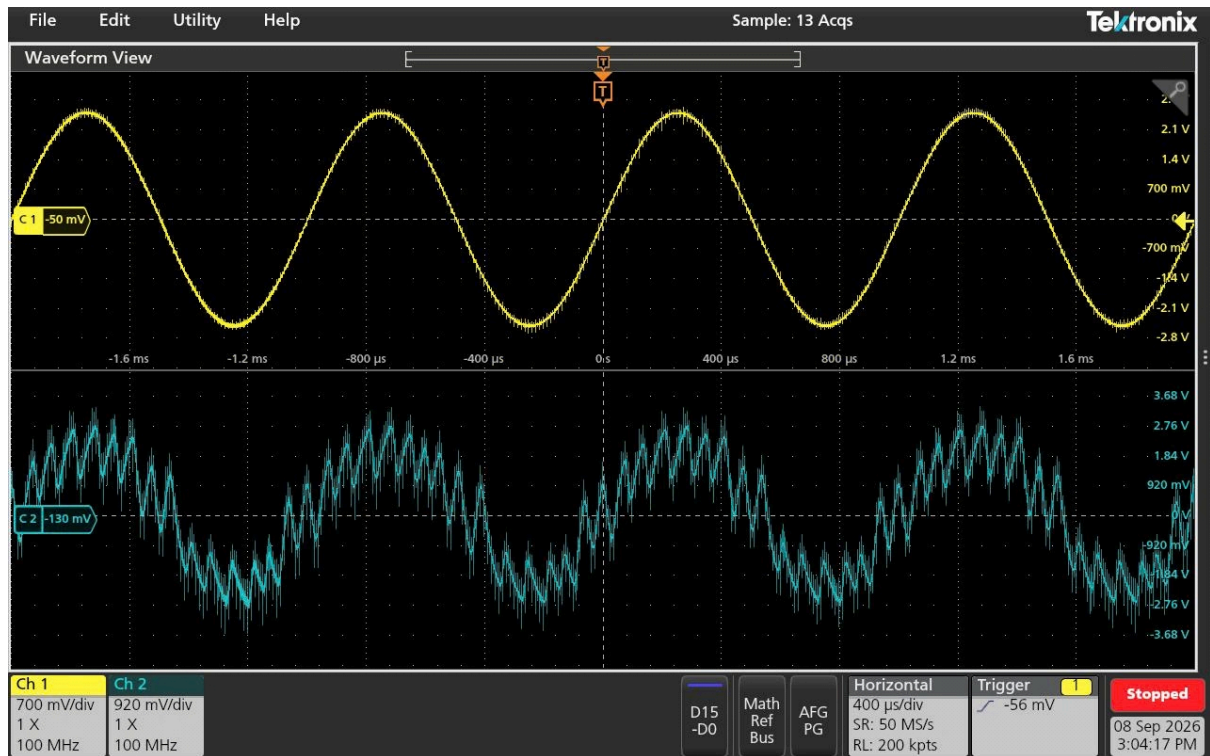


Fig 5.4 - Perfect Tracking. Optimal RC components yielded exactly $f_c = 4$ kHz. The step size Δ perfectly balances granular noise constraints while easily tracking the 1 kHz message signal's maximum slope.

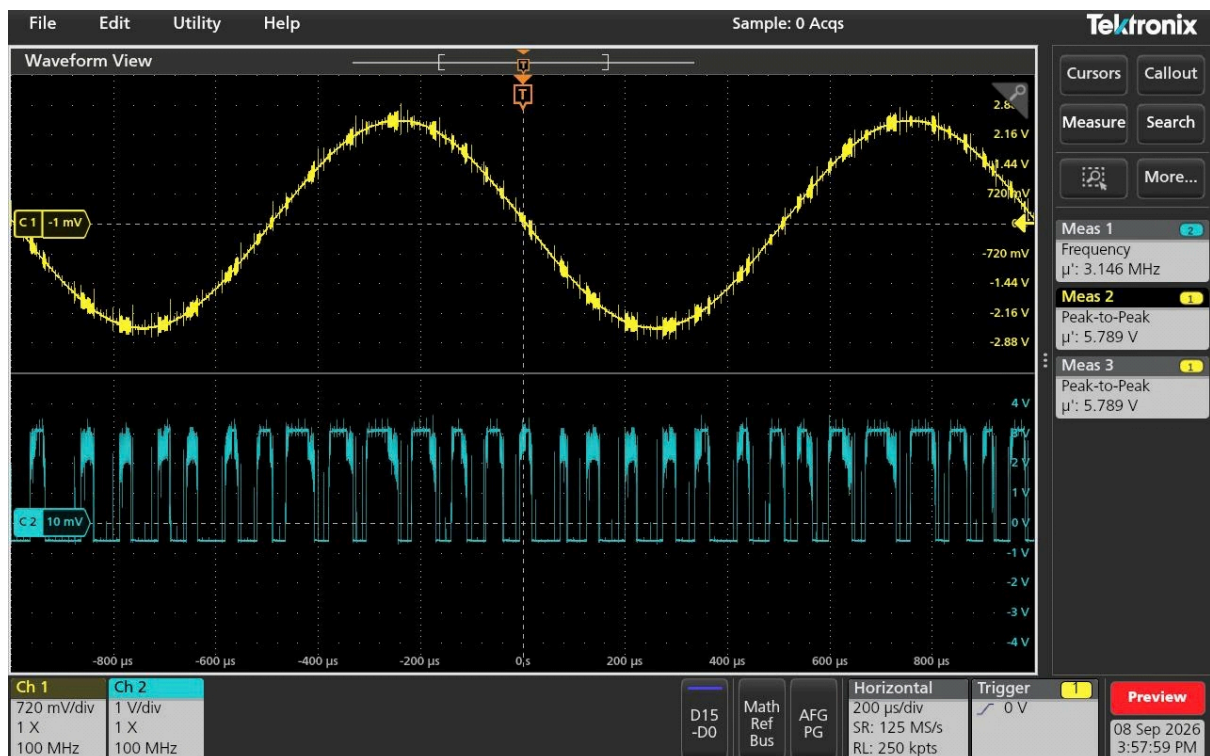


Fig 5.5 - Error Signal Detection. The raw, continuous-time output of the IC 710 differential comparator reflecting the immediate sign of the error $m(t) - \hat{m}(t)$.

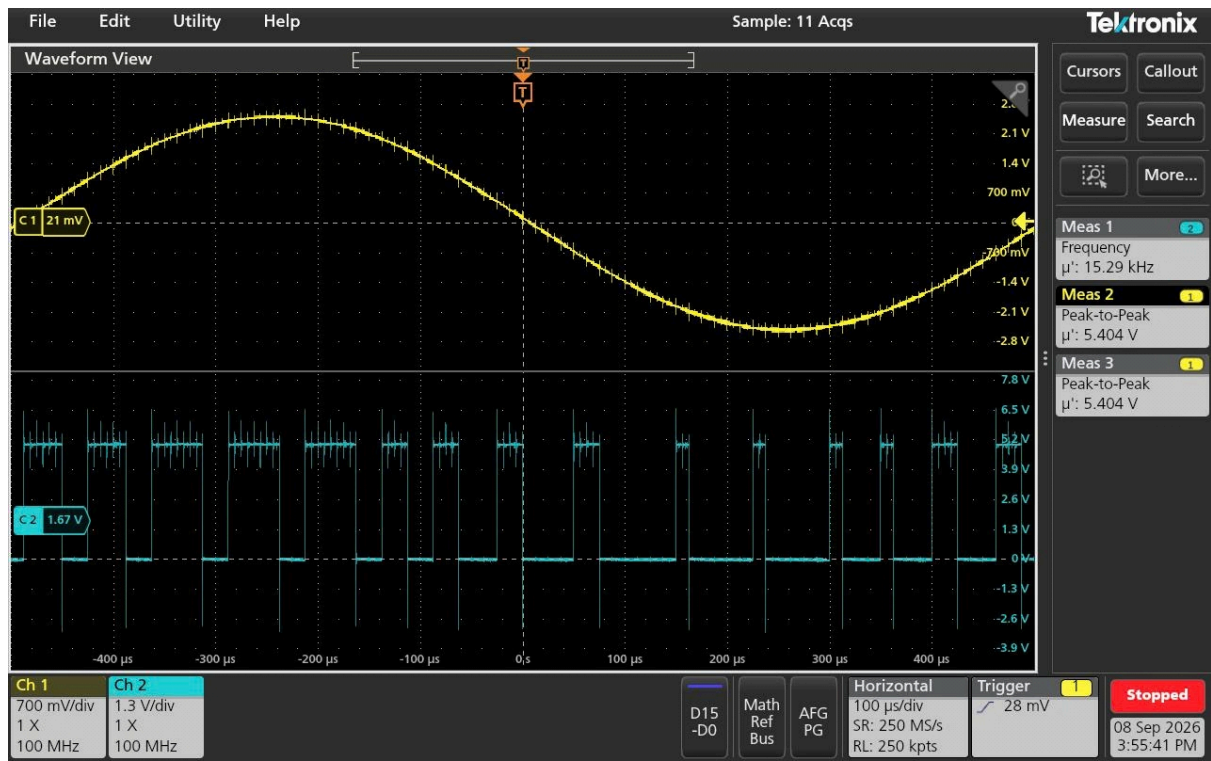


Fig 5.6 - Digital Pulse Stream. The quantized discrete-time output from the IC 7474 D-Flip Flop, strictly latched by the 80 kHz sampling clock.



Fig 5.7 - Demodulator Staircase Reconstruction. The intermediate waveform directly after the IC 356 active JFET integrator, perfectly regenerating the staircase $\hat{m}(t)$ before smoothing.



Fig 5.8 - Signal Overlay Comparisons. Oscilloscope overlays capturing the time-aligned original message $m(t)$ against the reconstructed waveforms, highlighting both the tracking fidelity and the expected LPF phase delay.

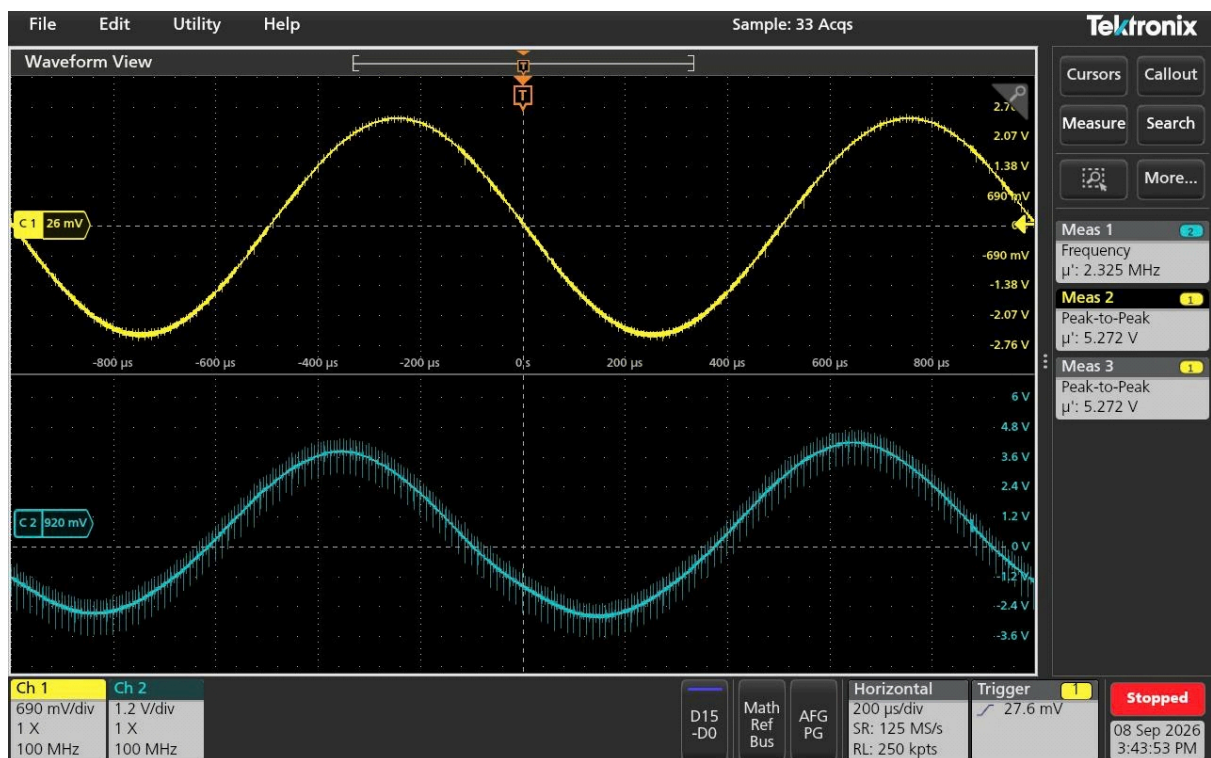


Fig 5.9 - Final Low-Pass Filtered Output. The fully recovered 1 kHz analog sine wave after stripping the high-frequency quantization noise via the final passive LPF.

5.2 MATLAB Digital Signal Processing & CSV Data Validation

The raw oscilloscope tracking data was exported as massive 250,000-point CSV arrays and imported into MATLAB for high-resolution analysis. We used MATLAB to unpack this data and visually verify both the macro-level tracking (phase delay) and the microscopic discrete steps of the 80 kHz sampling clock. The script (provided in **Appendix B**) dynamically parses the time vector to calculate exact physical time windows (e.g., a 2 ms window to capture two 1 kHz cycles, and a $150\mu\text{s}$ window to capture twelve 80 kHz clock steps), nullifying hardware DC offsets and exposing the granular operations that are normally heavily aliased on standard oscilloscope screens.

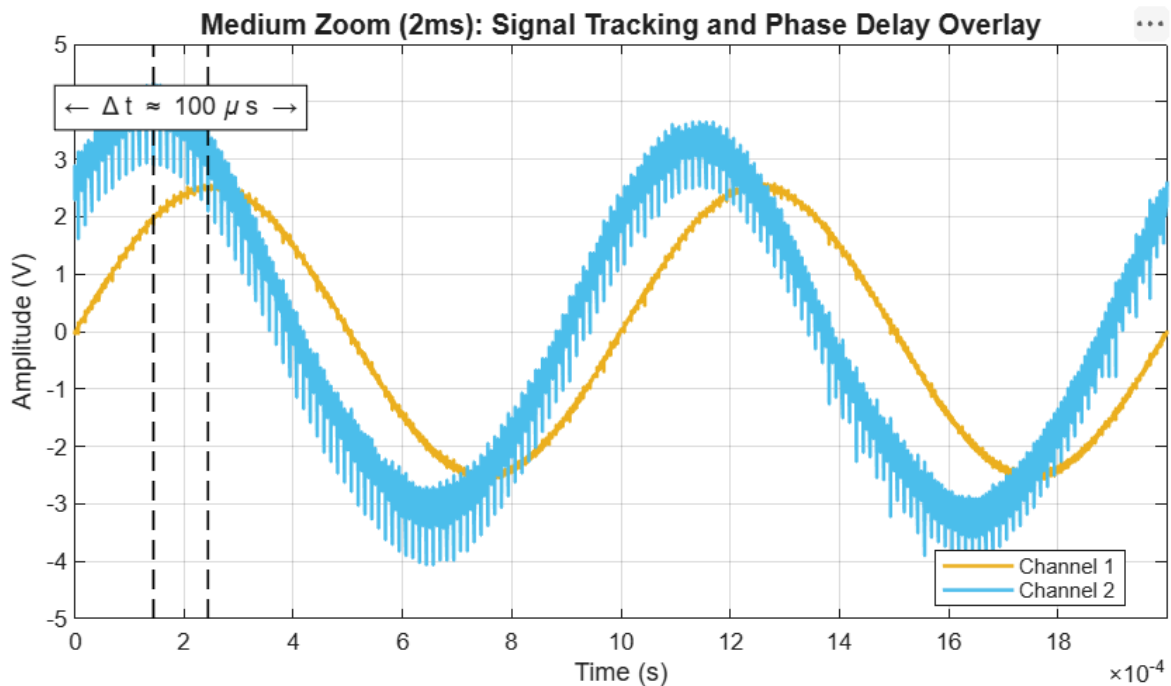


Fig 5.10 - MATLAB Phase Delay Analysis. A 2 ms time window capturing exactly two cycles of the 1 kHz signal, clearly revealing the group delay introduced by the hardware LPF.

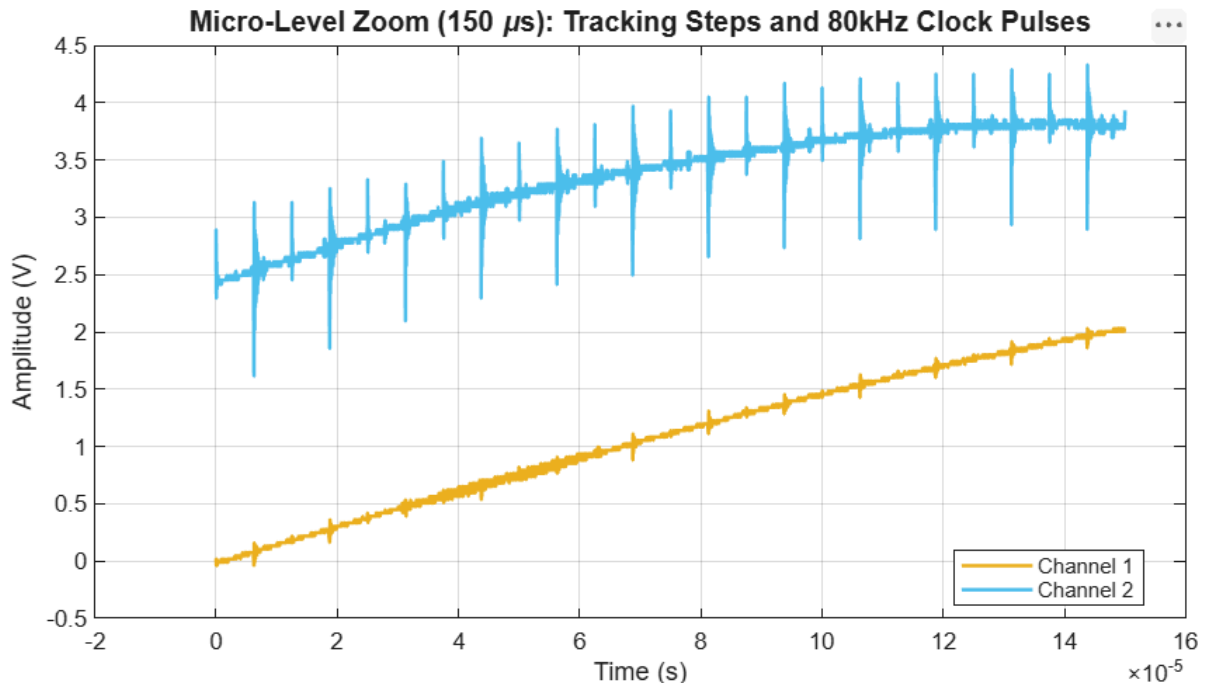


Fig 5.11 - MATLAB Micro-Level Quantization Steps. A 150μs window capturing roughly twelve 80 kHz sampling pulses, exposing the underlying discrete staircase integration steps.

6. INFERENCES & DISCUSSION

During the experiment, several critical design decisions were made to optimize system performance, deeply rooted in both theoretical limits and practical hardware physics:

1. Optimization of Integrator Time Constant (f_c): During the hardware implementation, the modulator's RC time constant was actively swept from a high equivalent cutoff (16 kHz) down to 1.5 kHz. At high f_c values (16 kHz and 10 kHz), the excessively small RC constant generated a massive step size Δ , resulting in severe granular noise that completely obscured the signal [2]. Conversely, lower f_c values (3 kHz and 2 kHz) created a massive RC constant and a tiny step size, causing textbook slope overload as the staircase failed to track the sine wave's rapid rise [3]. An optimal mathematical balance was established exactly at 4 kHz, successfully tracking the 1 kHz signal without either distortion [1].

2. Sampling Frequency (f_s) and Slope Overload Mitigation: Initial testing at $f_s=40$ kHz exhibited distinct slope overload on the steepest gradients of the message signal. Rather than increasing the step size (which would induce granular noise), the sampling frequency was doubled to 80 kHz. This satisfied the standard $f_s \geq 20f_m$ communication rule of thumb [2], doubling the modulator's maximum rate of rise and enabling the staircase to comfortably track steep slopes [3].

3. JFET Op-Amp (IC 356) for Demodulation Integrity: An IC 356 JFET operational amplifier was explicitly chosen for the demodulator's active integrator. Standard BJT op-amps exhibit input bias currents that bleed charge off the feedback integration capacitor, leading to a sagging, distorted staircase. The astronomically high input impedance and near-zero bias current of the JFET ensured that the capacitor held its integrated charge perfectly, yielding a highly stable and linear reconstruction [4].

4. Passive Integration in the Modulator Feedback Loop: Initial attempts to use an active op-amp integrator within the modulator's feedback loop resulted in severe DC drift and rail saturation, which broke the modulation cycle due to a lack of DC feedback stabilization [4]. The circuit was subsequently modified to use a naturally stable passive RC integrator, which is fully immune to DC runaway. The demodulator retained the active integrator because its subsequent passive LPF effectively blocked any baseline DC drift.

5. Voltage Buffer Isolation: A voltage follower (buffer) was inserted immediately *before* the modulator's passive RC integrator. The preceding level-shifting IC 741 operates as a comparator (in open-loop configuration), which can cause its output impedance to fluctuate dynamically. The buffer ensures the RC integration network is driven by a perfectly stiff, low-impedance $\pm 5\text{V}$ source, preserving the exact mathematical RC time constant [4].

6. Bipolar Pulse Generation via Level Shifting: The IC 7474 D-Flip Flop outputs standard unipolar TTL logic (0V and +5V). Because Delta Modulation mathematically requires the integrator to step down as well as up, applying unipolar logic would cause the staircase to saturate at the positive rail [1]. By using an IC 741 to compare the 0–5V signal against a precise 2.5V reference, the output was neatly shifted into a symmetric $\pm 5\text{V}$ bipolar pulse train, forcing proportional up and down integration steps [3].

7. Phase Delay and Signal Recovery Optimization: A prominent phase delay was observed in the final demodulated output relative to the original message signal (as clearly captured in the MATLAB Phase Delay Overlay, Fig 5.10). Theoretical analysis of our passive RC Low-Pass Filter ($R = 15\text{ k}\Omega$, $C = 10\text{ nF}$, yielding $f_c \approx 1061\text{ Hz}$) predicts a phase shift of $\varphi = \arctan(f_m / f_c) = \arctan(1000 / 1061) \approx 43.3^\circ$. For a 1 kHz message signal ($T = 1\text{ ms}$), this translates to a physical time delay of $\Delta t = (43.3^\circ / 360^\circ) \times 1\text{ ms} \approx 120\text{ }\mu\text{s}$. Visual inspection of the oscilloscope and MATLAB overlays perfectly confirms this $\sim 120\text{ }\mu\text{s}$ time gap between the signal peaks. This delay is an inherent mathematical property of causal filtering [1] and can only be eliminated by substituting a linear-phase active Bessel filter or using zero-phase digital DSP filtering.

7. CONCLUSION

The Delta Modulation and Demodulation scheme was comprehensively designed, debugged, and implemented in hardware. This experiment provided profound, hands-on insights into the physics of 1-bit quantization and oversampling. By actively tuning the sampling frequency and the integrator's RC time constants on the breadboard, we explicitly demonstrated the fundamental mathematical trade-off between slope overload and granular noise. The deployment of specialized components—such as JFET op-amps to prevent charge leakage, level-shifting comparators to achieve bipolar integration, and the critical shift from active to passive integration in the modulator loop—highlighted the severe analog constraints present in mixed-signal design. The final reconstructed signal accurately mirrored the original 1 kHz message, unequivocally proving the robustness and massive bandwidth efficiency of the Delta Modulation architecture.

8. REFERENCES

- [1] Lathi, B. P., & Ding, Z. (2009). *Modern Digital and Analog Communication Systems* (4th ed.). Oxford University Press. (Referenced heavily for A/D modulation advantages, PCM comparisons, and step size mathematics).
- [2] Haykin, S. (2001). *Communication Systems* (4th ed.). John Wiley & Sons. (Referenced for Nyquist Sampling Theory, 20x oversampling rule of thumb, and granular noise limits).
- [3] Proakis, J. G., & Salehi, M. (2008). *Digital Communications* (5th ed.). McGraw-Hill. (Referenced for detailed Comparator Logic, Differential PCM theory, Delta formulas, and Slope Overload conditions).
- [4] Lab Manual: *Digital Communication Laboratory (EC 39002)*, Spring 2025, IIT Kharagpur. (Base Implicit Reference).

APPENDIX A: IC PIN DIAGRAMS

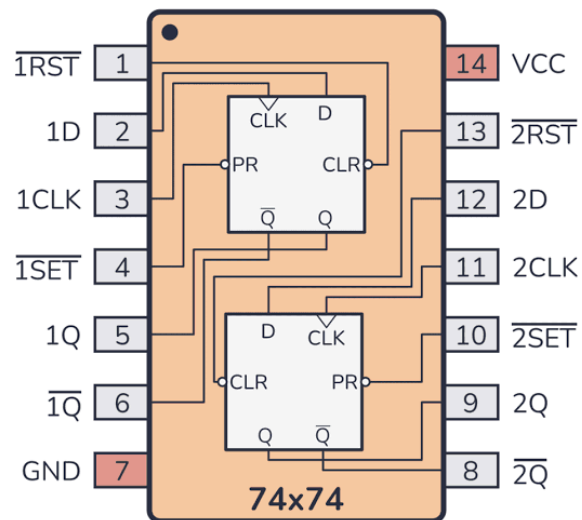


Fig A.1: Pin Diagram of IC7474

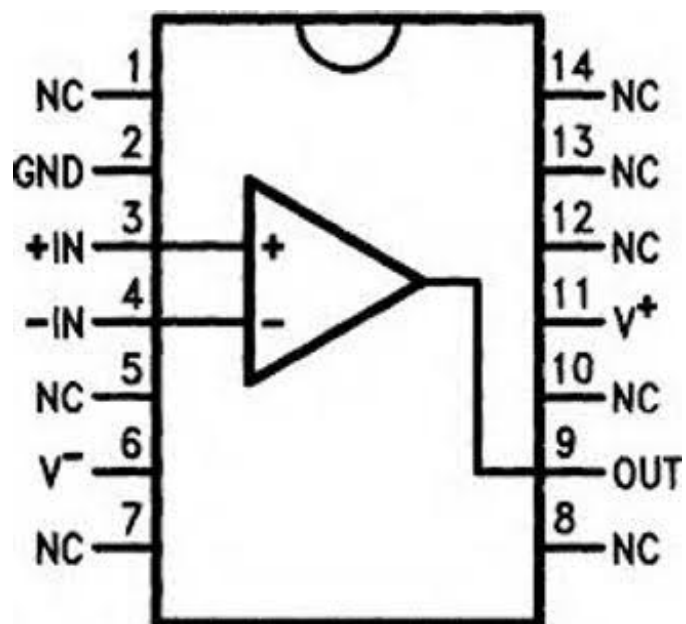


Fig A.2: Pin Diagram of IC 710

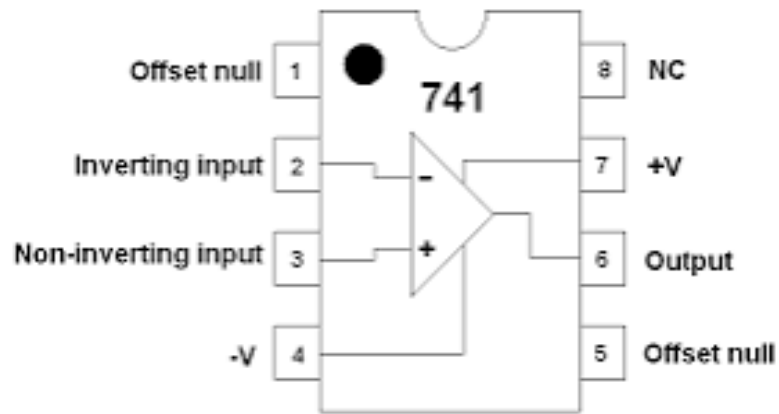


Fig A.3: Pin Diagram of IC741

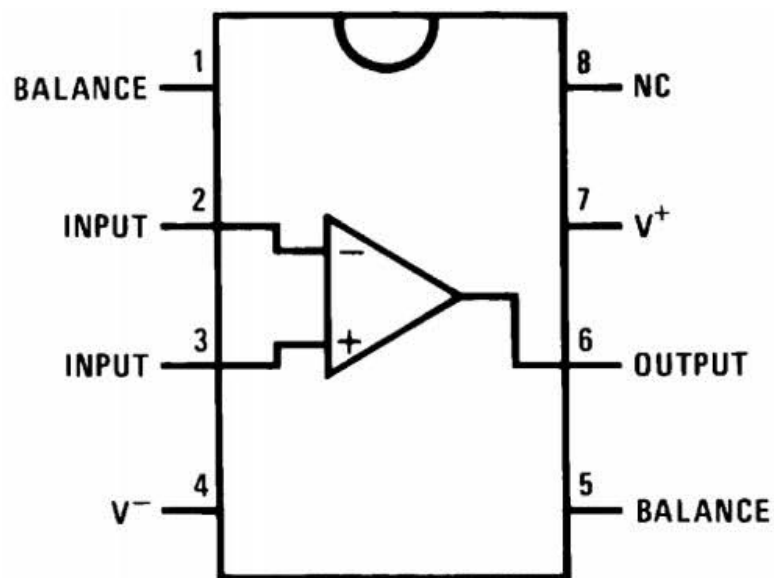


Fig A.4: Pin Diagram of IC 356

APPENDIX B: MATLAB SIMULATION CODE

```
clc;
clear all;
close all;

filename = 'demod_000_ALL.csv';
data = readmatrix(filename, 'NumHeaderLines', 14);
```

```
t = data(:, 1);
ch1 = data(:, 2);
ch2 = data(:, 3);

ch1 = ch1 - mean(ch1);
ch2 = ch2 - mean(ch2);

dt = t(2) - t(1);

mid = round(length(t)/2);

time_zoom = 2e-3;

pts1 = round(time_zoom / dt);

window1 = mid : min(mid + pts1, length(t));

half_len = round(length(window1)/2);
half_window = window1(1:half_len);

[~, p1] = max(ch1(half_window));
[~, p2] = max(ch2(half_window));

t1 = t(half_window(p1));
t2 = t(half_window(p2));

delay = abs(t2 - t1) * 1e6;

figure(1)

plot(t(window1), ch1(window1), 'LineWidth', 1.5);

hold on;

plot(t(window1), ch2(window1), 'LineWidth', 1.5);

yl = ylim;

plot([t1 t1], yl, '--k');
plot([t2 t2], yl, '--k');
```

```

txt = [' dt = ', num2str(round(delay)), ' us '];
text((t1+t2)/2, max(ch2(window1))*0.9, txt, 'HorizontalAlignment', 'center',
'BackgroundColor', 'w', 'EdgeColor', 'k');

title('Medium Zoom (2ms)');
xlabel('Time (s)');
ylabel('Voltage (V)');
legend('Message', 'Demodulated Output');
grid on;

time_micro = 150e-6;
pts2 = round(time_micro / dt);
window2 = mid : min(mid + pts2, length(t));

figure(2)
plot(t(window2), ch1(window2), 'LineWidth', 1.5);
hold on;
plot(t(window2), ch2(window2), 'LineWidth', 1.5);

title('Micro-Level Zoom (150 us)');
xlabel('Time (s)');
ylabel('Voltage (V)');
legend('Message', 'Demodulated Output');
grid on;

```